

CodeAlpha CGPA Calculator

Project Overview

This project is a CGPA Calculator developed in C++ as part of the CodeAlpha C++ Programming Internship.

The program allows students to enter course grades and credit hours, calculates total grade points and credits, and computes the final GPA/CGPA.

Features

- Input the number of courses.
- Enter grade points and credit hours for each course.
- Calculate total credits earned.
- Calculate total grade points.
- Compute GPA/CGPA automatically.
- Display course-wise information and final CGPA.

Technologies Used

- C++
- Standard Template Library (STL)
- Console-based User Interface

How It Works

1. Enter the number of courses.
2. Provide the grade point and credit hours for each course.
3. The program calculates:
4. Total Credits
5. Total Grade Points
6. GPA/CGPA
7. Results are displayed on the screen.

Sample Input

Number of Courses: 3

Course 1: Grade Point: 9 Credit Hours: 4

Course 2: Grade Point: 8 Credit Hours: 3

Course 3: Grade Point: 10 Credit Hours: 3

Sample Output

Course Details: Course 1 -> Grade: 9, Credits: 4 Course 2 -> Grade: 8, Credits: 3 Course 3 -> Grade: 10, Credits: 3

Total Credits: 10 Total Grade Points: 90

CGPA: 9.00

Learning Outcomes

- Understanding loops and arrays/vectors.
- Working with user input and output.
- Performing mathematical calculations in C++.
- Applying basic programming concepts to real-world problems.

Author

Monik K

Internship

CodeAlpha C++ Programming Internship